

Customer Behavior Analysis

Prepared for Alfido Tech — InternSpark Data Analytics Internship, Task 1

Dataset: Kaggle – E-Commerce Customer Behavior | Author: Adarsh Priyam

Executive Summary

This report analyzes 250,000 e-commerce transactions from 49,673 unique customers (Jan 2020 – Sep 2023) to segment customers, surface purchase patterns, and assess churn risk for Alfido Tech. Customers were segmented using RFM (Recency, Frequency, Monetary) scoring, cross-validated with K-Means clustering. Seven segments emerged, ranging from high-value **Champions** (17.6% of customers) to **Lost/Hibernating** customers (18.8%) who haven't purchased in over a year on average. Notably, the dataset's churn flag showed no meaningful correlation with purchase recency, frequency, spend, age, gender, or category — meaning churn here is likely driven by factors outside this dataset (e.g. service quality, pricing, delivery experience), which is itself an important finding. Five actionable recommendations are provided at the end of this report, covering win-back campaigns, loyalty programs, spend-based offers, churn data strategy, and category/margin strategy.

1. Data Overview & Cleaning

The raw dataset contained 250,000 rows and 13 columns (Customer ID, Purchase Date, Product Category, Price, Quantity, Total Purchase Amount, Payment Method, Age, Returns, Gender, and Churn). Data quality was high: no duplicate rows, and the only missing values were in the *Returns* column (47,596 rows), which were filled with 0 since a blank simply means no return was recorded for that order — not a data error. All purchase amounts and quantities were verified positive.

2. Methodology

RFM features were built per customer: Recency (days since last order, relative to the day after the dataset's latest date), Frequency (order count), and Monetary (total spend). Each was scored 1–5 by quintile and combined into 7 business-friendly segments using standard RFM thresholds. As a cross-check, **K-Means clustering** (k=4) was run independently on standardized RFM values; the resulting clusters lined up directionally with the RFM segments, confirming the segmentation reflects real structure in the data rather than arbitrary cutoffs.

3. Segment Profiles

Segment	Customers	% Base	Avg Recency (days)	Avg Frequency	Avg Monetary (amount)	Churn Rate
Champions	8,739	17.6%	61	7.7	21,785	20.0%
Loyal Customers	6,240	12.6%	149	6.9	17,440	20.0%
Big Spenders	3,590	7.2%	248	4.8	16,814	20.0%
At Risk	7,568	15.2%	408	6.1	16,032	19.0%
Potential Loyalists	8,794	17.7%	200	4.0	10,494	20.0%
New Customers	5,403	10.9%	68	3.2	8,802	21.0%
Lost / Hibernating	9,339	18.8%	580	2.6	6,488	20.0%

Champions and **Loyal Customers** together drive disproportionate value despite being ~30% of the base. **At Risk** customers (15.2%) still carry strong historical spend and frequency, making them high-priority win-back targets. **Lost/Hibernating** customers (18.8%) have the lowest engagement and lowest expected ROI on reactivation spend.

4. Visualizations

Customer Distribution Across Segments

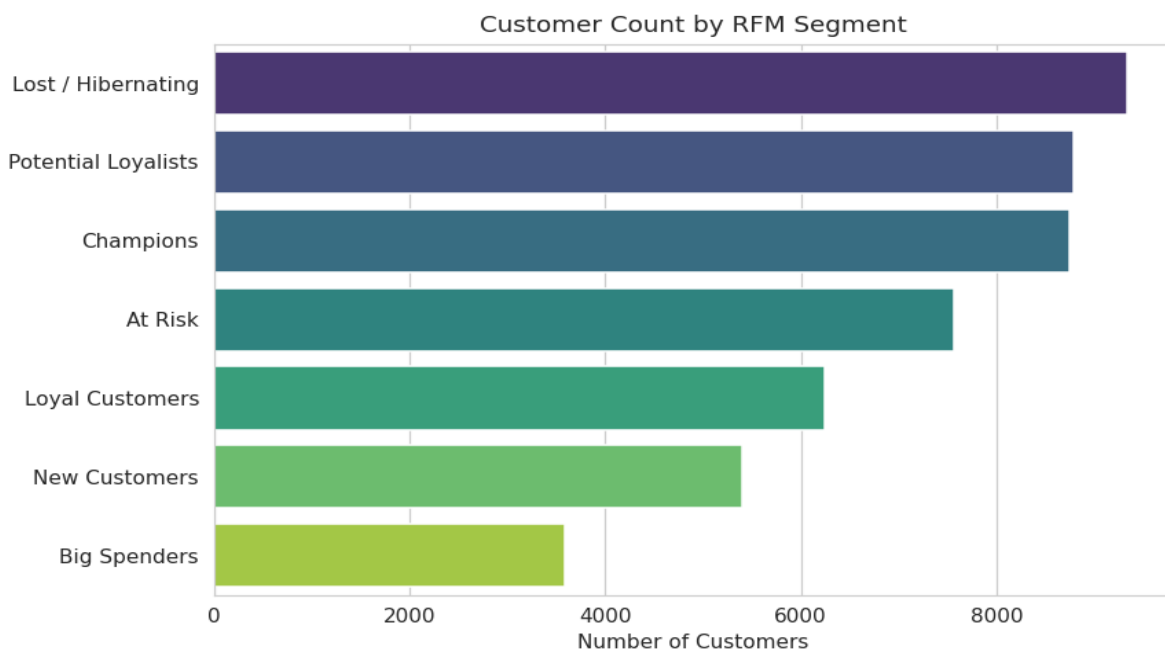


Figure 1. Lost/Hibernating and Potential Loyalists are the largest groups by headcount.

Spend Distribution by Segment

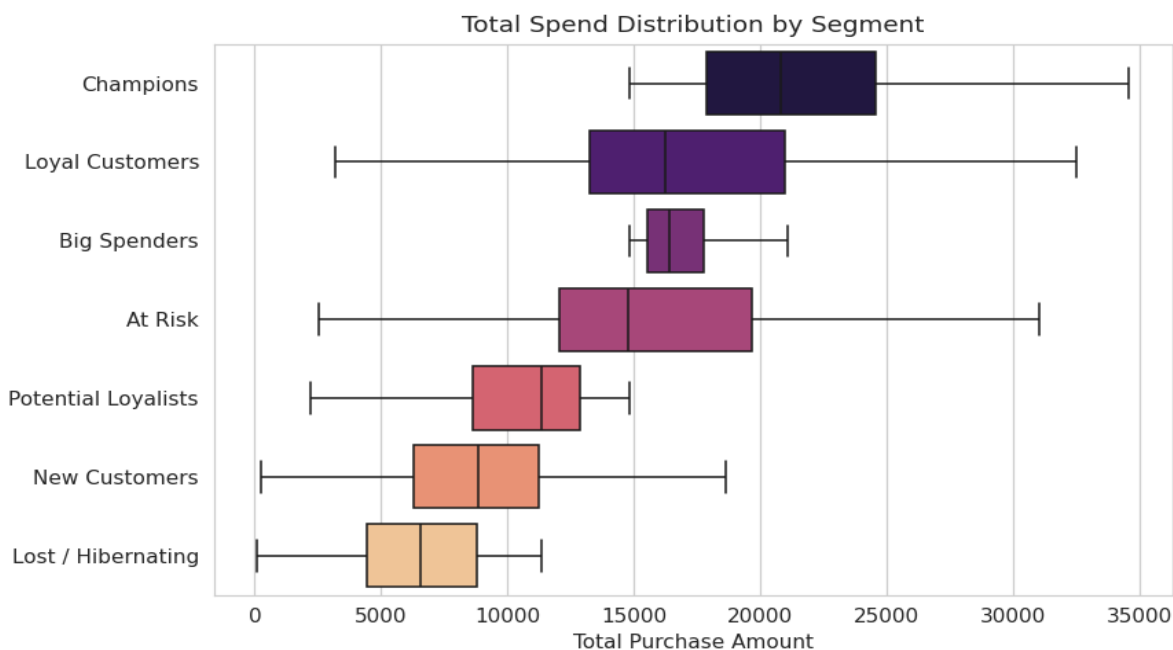


Figure 2. Champions and Loyal Customers show the highest and widest spread of total spend.

Churn Rate by Segment

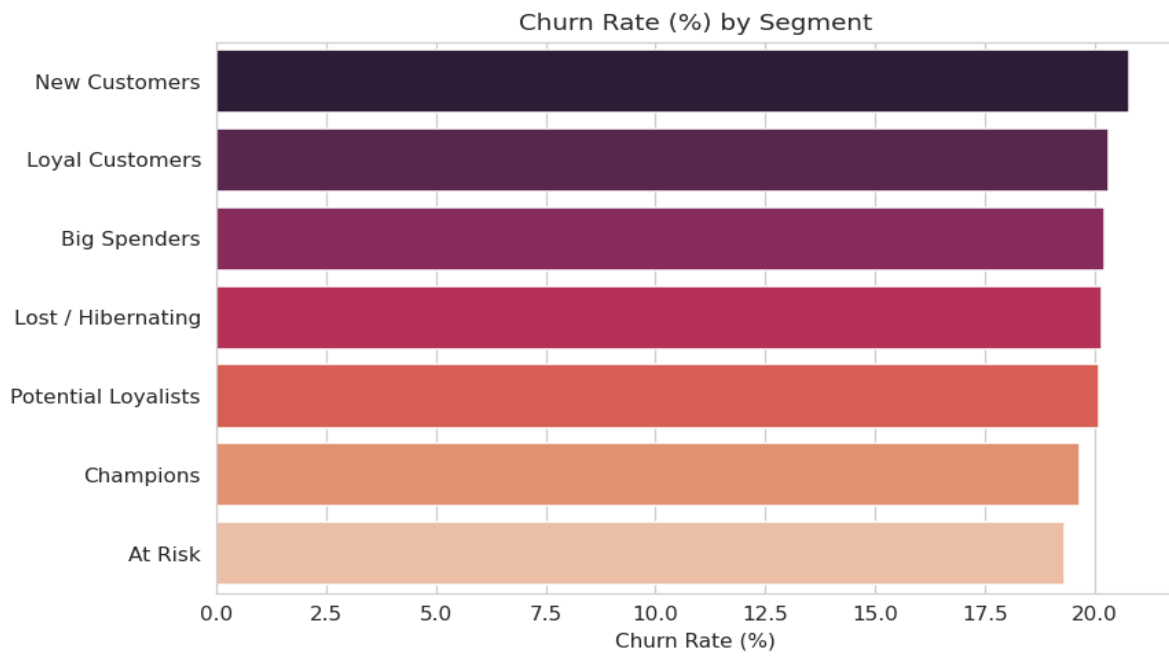


Figure 3. Churn rate is essentially flat (~20%) across every segment — a key finding (see Section 5).

Purchase Patterns by Category

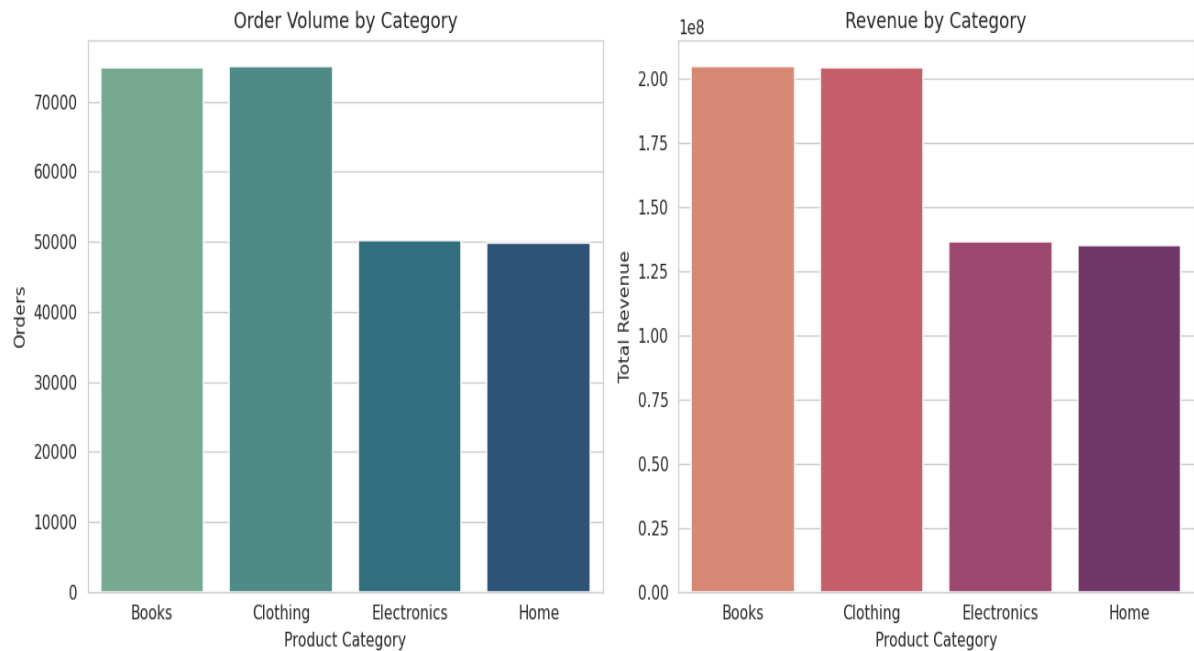


Figure 4. Clothing and Books lead in order volume and revenue; Electronics and Home trail despite similar AOV.

Retention Trend — Monthly Active Customers

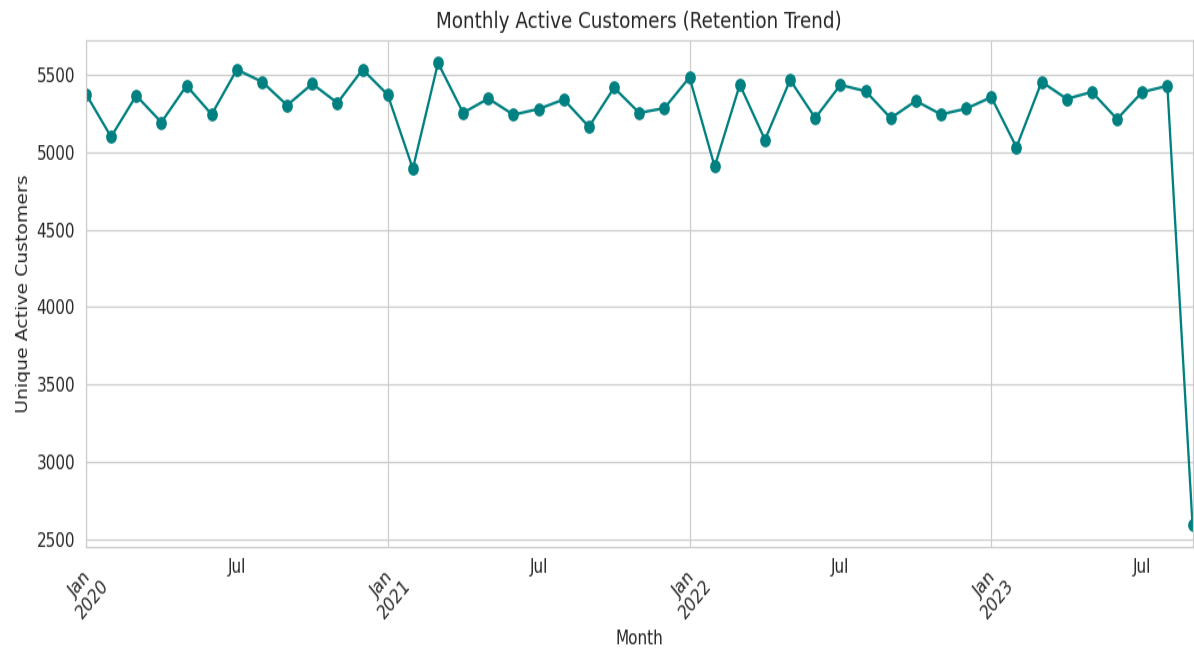


Figure 5. Monthly active customer counts over the dataset's full date range.

RFM Heatmap & Payment Methods

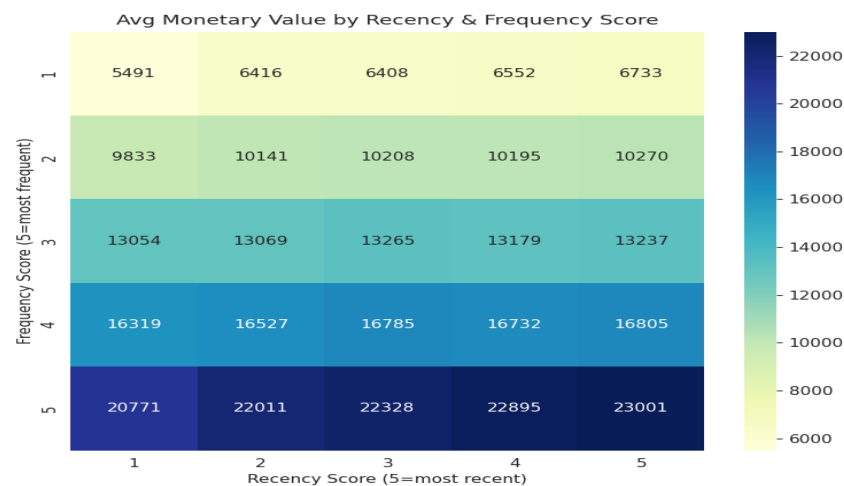


Figure 6. Average monetary value increases with both higher Recency and Frequency scores.

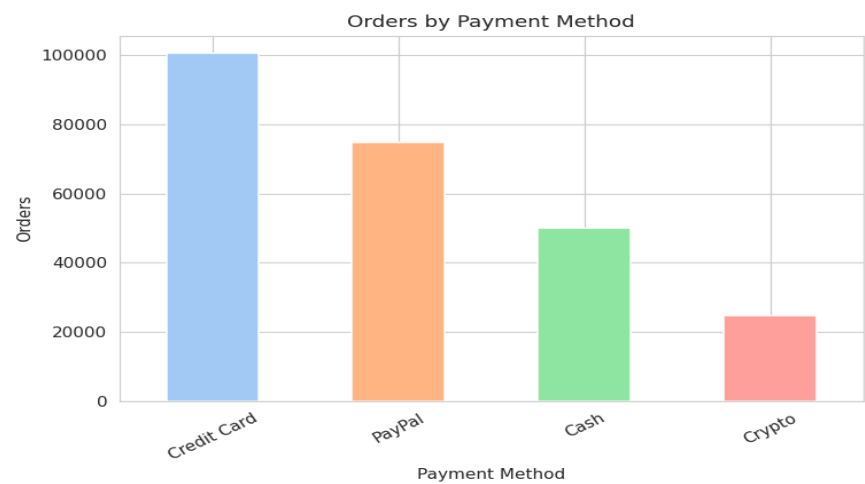


Figure 7. Order volume by payment method.

5. Key Finding — Churn is Not Explained by RFM Behavior

The dataset includes a Churn flag, but correlation analysis shows it is essentially independent of Recency, Frequency, Monetary value, Age, Gender, and Product Category — all correlations are below 0.01 in magnitude, and churn rate sits at roughly 20% across every RFM segment, from Champions to Lost/Hibernating. This means the common assumption that "inactive customers are the ones at risk" does not hold in this data. The real drivers of churn are most likely factors not captured here — support experience, delivery reliability, pricing versus competitors, or product quality. This is reported transparently rather than forcing a misleading 'high-risk segment' narrative.

6. Actionable Recommendations for Alfido Tech

1. Win back At-Risk customers

~7,500 customers (15.2%) have strong historical spend/frequency but haven't ordered in over a year on average. A targeted reactivation campaign (personalized discount, 'we miss you' email) is cheaper than acquiring a new customer of equal value.

2. Build a loyalty/VIP tier

Champions + Loyal Customers (~30% of the base) already drive outsized revenue. Early sale access, free-shipping thresholds, or referral perks can lock in retention and turn them into advocates.

3. Create spend-based offers for Big Spenders

This group has the highest average order value but lower frequency. Bundle deals and restock reminders timed to their purchase cycle can lift frequency without discounting, since they aren't price-sensitive.

4. Invest in churn data beyond RFM

Since churn doesn't correlate with purchasing behavior in this data, Alfido Tech should capture support tickets, delivery times, and review/NPS data to find the real churn drivers.

5. Prioritize Clothing & Books, audit Electronics margins

These categories lead order volume and revenue. Check whether Electronics' revenue-per-order justifies its cost relative to other categories, and consider cross-category bundles to lift average order value.

Full code, the executed notebook, and the underlying segment data accompany this report.